

Marc Recio

Data scientist who builds AI systems

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SUMMARY

Data scientist, plant breeder by training. I build language-model pipelines that curate the genomics literature, and I want to keep growing in machine learning, deep learning, data analysis and AI systems. I like problems that cross fields and solve them by building systems. On the side: Aevifex, my own method for working with AI agents.

SELECTED WORK

Human-in-the-loop gene curation

2025 – present

MSc data-science thesis (UOC) · with the TomsBioLab group, I2SysBio (CSIC – Universitat de València)

- A language-model pipeline that curates the literature for the gene catalogues of the PlantaeViz species: models read the papers, tested code assigns each gene's evidence level, and a curator holds two gates.
- Evaluated against a gold set curated by hand; the results are held until the paper and the defence.

A chemical screen for heat-stressed pollen

2024

MSc plant-breeding thesis (COMAV, UPV) · supervised by Dr. Ricardo Mir Moreno · graded 10/10

- 800 compounds screened on rapeseed pollen in a germination assay I designed, plated by an Opentrons OT-2 robot and counted in ImageJ: a single-well primary screen that left 8 candidates to retest.

Aevifex, my method for working with AI agents

2026

Own project · in daily use · private until its first release

- Rules, skills, hooks and a guard for Claude Code agents; the guard tested on 1,319 labelled tool calls, the checks on planted faults and controls.

EXPERIENCE

Internship, master's thesis

2026 – present

TomsBioLab, I2SysBio (CSIC – Universitat de València), Valencia

- Building the thesis's curation pipeline for the gene catalogues of the PlantaeViz species.

Research staff

May 2024 – Dec 2024

COMAV – Institute for Conservation and Improvement of Valencian Agrodiversity (UPV), Valencia

- Developed a protocol for obtaining transgenic rapeseed and ran in vitro microspore cultures (gametophytic development and androgenesis).
- PCR and sequencing for genetic analysis, plus histological analysis of salinity impact on reproductive development of *Cakile maritima*.
- Designed and carried out the statistical analyses behind the experimental results.

Laboratory technician (internship)

Jun 2023 – Nov 2023

Semillas Fitó, Cabrera de Mar, Barcelona

- In vitro procedures producing double haploids in pepper, cucumber and watermelon.
- DNA-extraction protocols in the molecular-biology department.

SKILLS

LLM pipelines	Structured extraction from papers, gold-set evaluation, provider-agnostic interfaces, rule-based decisions over model output.
Agents	Claude Code: rules, skills, hooks and guards, tested on labelled cases, planted faults and controls.
Machine learning	Supervised and unsupervised methods, evaluation on imbalanced data, PCA, t-SNE and UMAP; statistics in R.
Data & databases	SQL, PostgreSQL, schema design, ETL, web scraping, Power BI.
Programming	Python, R, SQL, Bash; Git, Docker, Linux.
Domain	Plant genomics and breeding, molecular biology, experimental design.
In progress	Deep learning and reinforcement learning.
Languages	Spanish (native), Valencian (Catalan, native), English (fluent, Cambridge B2 First), French (A2).

EDUCATION

MSc in Data Science	Feb 2025 – expected Jan 2027
Universitat Oberta de Catalunya (UOC) · top grade in Machine Learning	
MSc in Plant Genetic Breeding	Sep 2021 – Sep 2024
Universitat Politècnica de València (COMAV) · thesis graded 10/10	
BSc in Food Science and Technology	Sep 2017 – Sep 2021
Universitat Politècnica de València	

CERTIFICATES

• CS50's Introduction to Programming with Python — HarvardX (edX)	• Cambridge English: B2 First — Cambridge Assessment English
• Data Science: R Basics — HarvardX PH125.1x (edX)	• Valencian language, level C1 — JQCV